



Nutrition, Vitamins & Minerals

MBBS BIOCHEMISTRY · PAPER 1

Fat-Soluble Vitamins

Water-Soluble Vitamins

Folate Trap

Vitamin C & Collagen

Iron Metabolism

Calcium Homeostasis

One-Carbon Metabolism

Trace Minerals

PEM / Kwashiorkor

Balanced Diet

01 Fat-Soluble Vitamins

⚡ Fat-soluble = A, D, E, K. Stored in liver and adipose tissue. Absorbed with dietary fat (require bile salts). Toxicity possible with excess (especially A & D). NOT excreted in urine.

🧠 MNEMONIC

"**Fat Dads Eat Kale**" = **Fat-soluble: A (retinol), D (calciferol), E (tocopherol), K (koagulation)**

VITAMIN A (RETINOL)

★ PYQ 2025 – WALD'S VISUAL CYCLE

♦ **Forms:** Retinol (alcohol), Retinal (aldehyde), Retinoic acid (acid), β -Carotene (provitamin — plant source; $6\ \mu\text{g} = 1\ \mu\text{g}$ retinol)

♦ **Storage:** Liver as retinyl esters (stellate cells)

♦ **Transport:** Retinol-Binding Protein (RBP) + Transthyretin complex in blood

♦ **Biochemical roles:** Vision (retinal), Gene expression (retinoic acid $\rightarrow$ nuclear receptor RAR/RXR), Glycoprotein synthesis, Antioxidant (β -carotene)

Deficiency:

♦ **Night blindness** (first sign) — rod cell failure

♦ **Xerophthalmia** — dry eyes, Bitot's spots

♦ Keratomalacia — corneal ulceration $\rightarrow$ blindness

♦ Follicular hyperkeratosis (toad skin)

♦ Impaired immunity ($\downarrow$ IgA secretion)

Toxicity (Hypervitaminosis A):

♦ Acute: $\uparrow$ ICP, vomiting, headache

♦ Chronic: alopecia, hepatotoxicity, teratogenicity

► WALD'S VISUAL CYCLE – RHODOPSIN CYCLE (PYQ 2014-S, 2025)

Dark-adapted Rod Cell **11-cis Retinal** + Opsin → **Rhodopsin** (purple-red pigment, Scotopsin + 11-cis retinal)

↓ Light hits rhodopsin → photon absorbed

Bleaching reaction — isomerisation 11-cis Retinal → **all-trans Retinal** → Rhodopsin dissociates → Opsin released → conformational change → **nerve impulse generated**

↓ all-trans Retinal → all-trans Retinol (by retinol dehydrogenase, NADH)

Recovery in darkness (RPE cells) all-trans Retinol → all-trans Retinal → **11-cis Retinal** (by retinal isomerase) → combines with opsin → Rhodopsin regenerated Vitamin A is NOT consumed — it is recycled. But losses occur → dietary replenishment needed

🧐 NIGHT BLINDNESS – MOLECULAR BASIS

Vit A deficiency → insufficient 11-cis retinal → rhodopsin cannot be regenerated adequately after bleaching → rod cells fail to respond in dim light → **nyctalopia (night blindness)**. The cone cells (colour vision, bright light) use Iodopsin and are less affected initially. Bitot's spots = keratinised conjunctival cells, pathognomonic of Vit A deficiency.

VITAMIN D (CALCIFEROL) – THE HORMONE VITAMIN

★ PYQ 2019-S, 2023 – CALCIUM HOMEOSTASIS

► VITAMIN D ACTIVATION – STEP BY STEP

Step 1 — Skin (UV-B, 290-315 nm) **7-dehydrocholesterol** → UV-B → **Cholecalciferol (Vitamin D3)** Diet: D2 (ergocalciferol) from plants; D3 (cholecalciferol) from animals/fish liver oil

↓ Transported by Vitamin D-Binding Protein (DBP) to liver

Step 2 — Liver (25-hydroxylase) Cholecalciferol → **25-hydroxycholecalciferol (Calcidiol / 25-OH-D3)** Storage form — measured in blood to assess Vit D status

↓ Transported to kidney (PTH stimulates this step)

Step 3 — Kidney (1 α -hydroxylase) **KEY STEP** 25-OH-D3 → **1,25-dihydroxycholecalciferol (Calcitriol / 1,25-(OH)₂-D3)** ACTIVE form. Stimulated by: PTH, low Ca²⁺, low Pi. Inhibited by: high Ca²⁺, FGF-23

↓ Acts via nuclear receptor (VDR) — steroid hormone mechanism

Actions of Calcitriol Intestine: ↑ Ca²⁺ + Pi absorption (induces Calbindin D9k, TRPV6)
Bone: stimulates osteoblasts + mobilises Ca²⁺ (with PTH) Kidney: ↑ Ca²⁺ reabsorption
→ Net: ↑ serum Ca²⁺ and Pi

⚠️ Kidney failure → cannot perform 1 α -hydroxylation → active calcitriol ↓ → Ca²⁺ absorption ↓ → secondary hyperparathyroidism → renal osteodystrophy. Treatment: give active calcitriol (1,25-OH₂-D3) directly — NOT plain Vitamin D.

DEFICIENCY SYNDROME	AGE	FEATURES	BIOCHEMISTRY
Rickets	Children (growing bone)	Bow legs, Harrison's sulcus, rosary beads, frontal bossing, delayed dentition, craniotables	↓ Ca, ↓ Pi, ↑ ALP, ↑ PTH
Osteomalacia	Adults	Bone pain, muscle weakness, Looser's zones on X-ray (pseudofractures)	Same biochemistry as rickets
Hypervitaminosis D	Any	Hypercalcaemia, metastatic calcification, nephrolithiasis, anorexia	↑ Ca, ↑ Pi, ↑ 25-OH-D3

VITAMIN E (TOCOPHEROL)

PYQ 2012-S,
2022

- ◆ Most potent lipid-soluble **antioxidant**
- ◆ Protects polyunsaturated FA (PUFA) in membranes from lipid peroxidation
- ◆ Protects RBC membrane — prevents haemolysis
- ◆ Regenerated by Vitamin C (ascorbate)
- ◆ **Deficiency:** Haemolytic anaemia (especially in premature neonates), peripheral neuropathy, spinocerebellar ataxia, retinopathy
- ◆ Works synergistically with selenium (glutathione peroxidase)

VITAMIN K (PHYLLLOQUINONE)

★ PYQ 2014-S,
2019, 2023

- ◆ Essential cofactor for **γ-carboxylation** of glutamate residues in clotting factors
- ◆ **Factors dependent:** II (Prothrombin), VII, IX, X, Protein C, Protein S, Protein Z
- ◆ Mechanism: Vit K (hydroquinone form) donates electrons → γ-carboxylase adds CO₂ to Glu → γ-carboxyglutamate (Gla) → binds Ca²⁺ + phospholipid (essential for clot formation)
- ◆ Vit K epoxide formed → Vit K epoxide reductase regenerates Vit K (blocked by **Warfarin**)
- ◆ **K1** (Phylloquinone) = plants; **K2** (Menaquinone) = gut bacteria; **K3** (Menadione) = synthetic
- ◆ **Deficiency:** Bleeding tendency, prolonged PT/INR, **Haemorrhagic disease of newborn (HDN)** — neonates have sterile gut, no K2 synthesis

02

B-Complex Vitamins

- ⚡ B-vitamins are water-soluble coenzymes. Most are NOT stored significantly → daily dietary requirement. Excreted in urine (toxicity rare). Exception: B12 (stored in liver for 3–5 years).

VITAMIN	ACTIVE COENZYME	KEY REACTION(S)	DEFICIENCY DISEASE	PYQ
B1 — Thiamine	Thiamine Pyrophosphate (TPP)	PDH complex (pyruvate→AcCoA), α-KG dehydrogenase (TCA), Transketolase (HMP shunt)	Beriberi (wet = cardiac; dry = neuropathy), Wernicke-Korsakoff (alcoholics)	2013, 2021
B2 — Riboflavin	FAD, FMN	ETC Complex I & II, β-oxidation (acyl-CoA dehydrogenase), PDH (E3)	Ariboflavinosis: glossitis, angular stomatitis, corneal vascularisation, photophobia	2016
B3 — Niacin (PP factor)	NAD ⁺ , NADP ⁺	Redox reactions in glycolysis, TCA, ETC, HMP shunt, fatty acid synthesis	Pellagra: 3Ds — Dermatitis (photosensitive), Diarrhoea, Dementia. 4th D = Death	2013, 2016-S
B5 — Pantothenic acid	Coenzyme A (CoA)	Acyl carrier for all CoA-linked reactions (Acetyl-CoA, Succinyl-CoA, Malonyl-CoA)	Rare — burning feet syndrome	—
B6 — Pyridoxine	Pyridoxal Phosphate (PLP)	Transamination, decarboxylation of amino acids, glycogen phosphorylase, haem synthesis (ALA synthase)	Peripheral neuropathy, convulsions (↓ GABA synthesis), sideroblastic anaemia	2017-S
B7 — Biotin	Biocytin (protein-bound)	Carboxylation reactions: Pyruvate carboxylase, Acetyl-CoA carboxylase (RLS of FA synthesis), Propionyl-CoA carboxylase	Rare: Dermatitis, alopecia, glossitis. Avidin in raw egg whites binds biotin (competitive)	2015-S
B9 — Folic Acid	Tetrahydrofolate (THF)	One-carbon transfer reactions (thymidylate synthesis, purine synthesis, methionine synthesis)	Megaloblastic anaemia, Neural tube defects (NTD) in pregnancy	2011-S, 2016-S, 2018, 2025
B12 — Cobalamin	Methylcobalamin, Adenosylcobalamin	Methionine synthesis (with folate), Methylmalonyl-CoA → Succinyl-CoA (odd-chain FA)	Megaloblastic anaemia, Subacute combined degeneration of spinal cord (SACD), Folate trap	2022, 2024, 2025

NIACIN — SYNTHESIS FROM TRYPTOPHAN

PYQ 2013, 2016-S — PELLAGRA

- ◆ Niacin can be synthesised endogenously from **Tryptophan** (60 mg Trp = 1 mg niacin equivalent)
- ◆ Requires Vitamins B2, B6, and iron for the conversion
- ◆ Hartnup disease → defective neutral amino acid transporter → ↓ Trp absorption → pellagra-like syndrome
- ◆ Carcinoid syndrome → excess Trp diverted to serotonin synthesis → pellagra

PELLAGRA — 4 DS

Dermatitis: photosensitive, symmetric, Casal's necklace around neck

Diarrhoea: malabsorption, mucosal atrophy

Dementia: depression, confusion, eventually dementia

Death (if untreated)

Classic association: maize-dependent diet (low Trp + niacin bound as niacytin — not bioavailable)

PYRIDOXAL PHOSPHATE (PLP) — MASTER AMINO ACID COENZYME

PYQ 2017-S

- ⚡ PLP participates in **all** reactions involving amino acid nitrogen: transamination, decarboxylation, racemisation, β -elimination, γ -replacement. It forms a Schiff base with the amino acid's α -amino group at the enzyme active site.

REACTION TYPE	EXAMPLE	CLINICAL RELEVANCE
Transamination	Asp + α -KG $\rightleftharpoons$ OAA + Glu (AST); Ala + α -KG $\rightleftharpoons$ Pyruvate + Glu (ALT)	AST/ALT elevated in liver disease — both need PLP
Decarboxylation	DOPA → Dopamine; 5-HTP → Serotonin; Glutamate → GABA; Histidine → Histamine	B6 deficiency → ↓ GABA → seizures in neonates
ALA Synthase	Glycine + Succinyl-CoA → δ -ALA (rate-limiting step of haem synthesis)	B6 deficiency → sideroblastic anaemia
Glycogen Phosphorylase	Glycogen breakdown in muscle	PLP covalently bound — structural role
Cystathionine β-synthase	Homocysteine + Serine → Cystathionine (transsulphuration)	B6 deficiency → homocysteinaemia → atherosclerosis

03 Vitamin C & Collagen

VITAMIN C (ASCORBIC ACID)

★ PYQ 2018-S, 2021, 2017-S

Biochemical Functions:

- ◆ **Collagen synthesis:** Cofactor for Prolyl hydroxylase AND Lysyl hydroxylase → hydroxyproline + hydroxylysine formation → essential for triple helix stability
- ◆ **Antioxidant:** Scavenges ROS ($O_2^{\bullet-}$, $OH^{\bullet}$, H_2O_2). Regenerates Vitamin E from tocopheryl radical
- ◆ **Iron absorption:** Reduces $Fe^{3+} \rightarrow Fe^{2+}$ (absorbed form) in intestinal lumen → ↑ non-haem iron absorption (~3-fold)
- ◆ **Norepinephrine synthesis:** Cofactor for Dopamine β-hydroxylase
- ◆ **Hydroxylation of drugs/steroids** (cytochrome P450)
- ◆ Carnitine synthesis (two hydroxylation steps)
- ◆ Tyrosine metabolism

Scurvy — Deficiency:

- ◆ **Perifollicular haemorrhage** (corkscrew hairs) — pathognomonic
- ◆ Bleeding gums, gingivitis, tooth loosening
- ◆ Poor wound healing
- ◆ Subperiosteal haemorrhage (children — painful)
- ◆ Anaemia (haemolytic + microcytic)
- ◆ Delayed bone healing — Trümmerfeld zone (Barlow's disease in infants)
- ◆ *Mechanism: defective collagen → weakened vessel walls + connective tissue*

SCURVY SYMPTOMS

"Bleeding Gums, Wound Healing, Perifollicular = BaWP" — Barlow's, Weakness, Perifollicular haemorrhage

04 Folate & B12 — Folate Trap

- ⚡ This is the highest-yield biochemistry topic in this module. The folate trap explains why B12 deficiency causes megaloblastic anaemia AND neurological damage while folate deficiency causes ONLY megaloblastic anaemia (no neuro).

► FOLATE METABOLISM & THE METHYL FOLATE TRAP

Dietary Folate (Folic Acid) Dietary polyglutamate → Deconjugated → **Folic acid (PteGlu)** → Reduced to **Dihydrofolate (DHF)** by DHFR → Further reduced to **Tetrahydrofolate (THF)** — active form

↓ THF accepts one-carbon units → methyleneTHF → methylTHF

Key Reaction — Methionine Synthase (requires B12) **Methyl-THF** + Homocysteine → **THF** + Methionine This reaction REQUIRES Vitamin B12 (as methylcobalamin) to transfer the methyl group

↓ If B12 deficient → methyl-THF CANNOT donate its methyl group → accumulates

THE FOLATE TRAP (B12 deficiency) All cellular folate becomes "trapped" as **methyl-THF** — the monoglutamate form cannot be re-polyglutamated and retained inside cells → No free THF available → **cannot synthesise thymidylate (dTMP)** → DNA synthesis impaired → megaloblastic anaemia (same as folate deficiency) But: homocysteine accumulates → demyelination → **NEUROLOGICAL DAMAGE** (B12-specific, not seen in folate deficiency alone)

↓ Free THF used in thymidylate synthesis

Thymidylate Synthase Reaction (DNA synthesis) dUMP + **Methylene-THF** → dTMP + DHF DHF → THF (by DHFR). Blocked by: Methotrexate (DHFR inhibitor), 5-FU (thymidylate synthase inhibitor)

FEATURE	FOLATE DEFICIENCY	B12 DEFICIENCY
Megaloblastic anaemia	Yes	Yes (via folate trap)
Neurological damage (SACD)	No	Yes — demyelination of dorsal + lateral columns
Serum folate	↓	↑ (trapped as methyl-THF)
Homocysteine	↑	↑↑
Methylmalonic acid (MMA)	Normal	↑↑ (diagnostic for B12 deficiency)
Giving folic acid to B12-deficient patient	—	Corrects anaemia BUT worsens neuro damage — DANGER
Intrinsic Factor (IF)	Not needed	Required for B12 absorption — deficiency → pernicious anaemia



Never give folate alone to a B12-deficient patient without giving B12. It corrects the anaemia (mask) but the neurological degeneration (subacute combined degeneration) continues and worsens undetected.

🧬 NEURAL TUBE DEFECTS & FOLATE IN PREGNANCY

Folate required for de novo purine and thymidylate synthesis → DNA synthesis for neural tube closure (days 21–28 of gestation — before most women know they're pregnant). Periconceptional folic acid supplementation (**400 µg/day** before conception and in first trimester) reduces NTD risk by ~70%. NTDs: anencephaly, spina bifida, meningomyelocele.

05

One-Carbon Metabolism & SAM

- ⚡ One-carbon metabolism = transfer of single carbon units between molecules. Folate and B12 are the key carriers. SAM is the universal methyl donor for most biological methylation reactions.

► SAM CYCLE – ONE-CARBON METABOLISM HUB

Methionine Activation (ATP-dependent) Methionine + ATP → **S-Adenosylmethionine (SAM)** + PPi + Pi Catalysed by Methionine adenosyltransferase. SAM = universal methyl donor

↓ SAM donates CH₃ to any acceptor (DNA, RNA, proteins, phospholipids, neurotransmitters)

Methylation Reactions by SAM SAM → **S-Adenosylhomocysteine (SAH)** after donating methyl group Examples: Norepinephrine → Epinephrine (COMT); Guanidinoacetate → Creatine; DNA CpG methylation (epigenetics); Phosphatidylethanolamine → Phosphatidylcholine; Histamine → N-methylhistamine

↓ SAH hydrolysed → Homocysteine + Adenosine

Homocysteine Fate **Route 1 (Remethylation)**: Homocysteine + Methyl-THF → Methionine (needs B12 + Methionine synthase) — salvage pathway **Route 2 (Transsulphuration)**: Homocysteine + Serine → Cystathionine → Cysteine (needs PLP/B6) Cysteine → Glutathione (GSH), Taurine, Sulphate.

HOMOCYSTEINAEMIA

PYQ 2024 – B12 DEFICIENCY CASE

Causes of elevated homocysteine:

- ◆ B12 deficiency (↓ remethylation)
- ◆ Folate deficiency (↓ methyl-THF supply)
- ◆ B6 deficiency (↓ transsulphuration)
- ◆ Homocystinuria — CBS deficiency (autosomal recessive)
- ◆ MTHFR mutation (C677T polymorphism — most common)

Clinical consequences:

- ◆ Accelerated atherosclerosis (endothelial damage)
- ◆ Arterial and venous thrombosis
- ◆ Homocystinuria: lens dislocation (ectopia lentis), Marfanoid habitus, intellectual disability, osteoporosis, thromboembolism
- ◆ Independent CVD risk factor

ONE-CARBON UNITS – OXIDATION STATES

PYQ 2014-S

OXIDATION STATE	CARBON UNIT	THF DERIVATIVE	MAIN USE
Most oxidised	Formate (–CHO)	N10-formyl-THF	Purine ring synthesis (C2, C8)
Intermediate	Formaldehyde (–CH ₂ –)	N5,N10-methylene-THF	dTMP synthesis (thymidylate), serine↔glycine
Most reduced	Methanol (–CH ₃)	N5-methyl-THF	Methionine synthesis (remethylation)

06 Iron Metabolism

⚡ Iron is the most commonly deficient micronutrient worldwide. The body has no regulated excretory mechanism for iron — all regulation happens at the level of ABSORPTION. Total body iron ~4g in males, ~2.5g in females.

► IRON ABSORPTION, TRANSPORT & STORAGE

Dietary Sources & Forms **Haem iron** (myoglobin, Hb from meat): absorbed directly as intact haem via HCP1 receptor → ~25% absorption, not affected by dietary factors
Non-haem iron (ferric Fe³⁺ from plants, cereals): must be reduced to Fe²⁺ first → absorbed by DMT-1 (divalent metal transporter) → lower bioavailability (~5%) Vitamin C (ascorbate) reduces Fe³⁺ → Fe²⁺ → ↑ non-haem absorption 3-fold

↓ Inside enterocyte — iron can be stored as Ferritin or exported

Export from Enterocyte — Basolateral Side Fe²⁺ exported by **Ferroportin** (only iron exporter in body) → Oxidised to Fe³⁺ by **Hephaestin** (ferroxidase, copper-dependent) → Binds **Apotransferrin** → **Transferrin** (carries 2 Fe³⁺ atoms, β₁-globulin)

↓ Transferrin delivers iron to cells via Transferrin Receptor (TfR)

Cellular Uptake — Receptor-mediated Endocytosis Transferrin-TfR complex → endosome (acidified) → Fe³⁺ released → reduced to Fe²⁺ by Steap3 → exits endosome via DMT-1 → used for haem/Fe-S cluster synthesis or stored as Ferritin

↓ Excess stored; regulation by Hepcidin

Iron Storage — Ferritin & Haemosiderin **Ferritin**: 24 subunits, stores up to 4500 Fe atoms as Fe(OH)₃ core. Serum ferritin = best indicator of iron stores (1 ng/mL ≈ 8 mg stored iron) **Haemosiderin**: Degraded ferritin, insoluble, stains with Perl's Prussian blue. Seen in iron overload.

HEPCIDIN — MASTER REGULATOR OF IRON

★ HIGH YIELD

⚡ Hepcidin = 25-amino acid peptide hormone made by LIVER. It is the key iron hormone that controls the amount of iron entering the circulation.

When iron/inflammation HIGH → Hepcidin ↑:

- ◆ Hepcidin binds Ferroportin → internalisation and degradation of Ferroportin
- ◆ No Ferroportin → iron TRAPPED in enterocytes + macrophages
- ◆ Iron absorption ↓, iron recycling ↓
- ◆ Serum iron ↓

When iron LOW / hypoxia / anaemia → Hepcidin ↓:

- ◆ Ferroportin expressed on enterocyte surface
- ◆ Iron freely exported into plasma
- ◆ Iron absorption ↑
- ◆ Serum iron ↑

ANAEMIA OF CHRONIC DISEASE – HEPCIDIN MECHANISM

Chronic infection/inflammation → IL-6 → liver produces ↑ Hepcidin → Ferroportin degraded → iron trapped in macrophages → iron not available for erythropoiesis → **normocytic/microcytic anaemia** with normal-to-high iron stores (serum ferritin ↑, serum iron ↓, TIBC ↓ — distinguishes from iron deficiency anaemia).

IRON LAB TESTS – MBBS EXAM MUST-KNOW

TEST	NORMAL	IRON DEFICIENCY ANAEMIA	ANAEMIA OF CHRONIC DISEASE	HAEMOCHROMATOSIS
Serum Iron	60–170 µg/dL	↓	↓	↑↑
TIBC (Transferrin)	250–370 µg/dL	↑↑	↓ or N	↓
% Saturation	20–50%	↓↓ (<15%)	↓	↑↑ (>60%)
Serum Ferritin	15–200 ng/mL	↓↓ (first to fall)	↑ or N	↑↑↑
Hepcidin	—	↓	↑↑	↓ (mutant HFE gene)

HAEMOCHROMATOSIS – IRON OVERLOAD

Hereditary haemochromatosis: mutation in **HFE gene** (Chr 6) → defective regulation → Hepcidin low despite iron overload → Ferroportin constitutively active → iron overabsorbed → deposits in liver (cirrhosis), pancreas (bronze diabetes), heart (cardiomyopathy), skin (bronze pigmentation), gonads (hypogonadism). Triad: cirrhosis + DM + skin pigmentation = "Bronze diabetes". Treatment: Phlebotomy.

07

Calcium Homeostasis & Vitamin D

⚡ Normal serum calcium: 8.5–10.5 mg/dL. 50% ionised (active), 40% albumin-bound, 10% complexed. Regulated by: PTH, Calcitriol (1,25-OH₂-D₃), Calcitonin.

► CALCIUM HOMEOSTASIS – 3-HORMONE SYSTEM

Trigger: Serum Ca²⁺ falls below normal Parathyroid glands sense ↓ Ca²⁺ via CaSR (calcium-sensing receptor)

↓ PTH secreted within minutes

PTH Actions (rapid, within minutes to hours) **Bone:** Osteoclast stimulation (via osteoblast → RANKL) → bone resorption → Ca²⁺ + Pi released **Kidney:** ↑ Ca²⁺ reabsorption (distal tubule), ↑ Pi excretion (phosphaturia), ↑ 1α-hydroxylase → calcitriol production Net: ↑ serum Ca²⁺, ↓ serum Pi

↓ PTH also stimulates calcitriol production (slower, hours-days)

Calcitriol Actions (slower, genomic) **Intestine:** ↑ Ca²⁺ absorption (Calbindin D9k), ↑ Pi absorption (NaPi-IIb) **Bone:** facilitates mineralisation (with adequate Ca/Pi) **Kidney:** ↑ Ca²⁺ reabsorption Net: ↑↑ serum Ca²⁺ and Pi

↓ When Ca²⁺ returns to normal → thyroid C-cells respond

Calcitonin (from thyroid C-cells) — when Ca²⁺ HIGH Inhibits osteoclasts → ↓ bone resorption ↑ Renal Ca²⁺ excretion Net: ↓ serum Ca²⁺ Less important in adults; marker for medullary thyroid carcinoma (MTC)

HYPOCALCAEMIA

Causes: Hypoparathyroidism, Vit D deficiency, CKD, hypomagnesaemia (↓ PTH secretion + resistance).

Clinical: Tetany (Chvostek's sign — facial twitch; Trousseau's sign — carpopedal spasm with BP cuff), convulsions, QT prolongation, cataracts (chronic).

Biochemistry: ↓ Ca, ↑ PTH (secondary), ↑ ALP (if Vit D deficiency)

HYPERCALCAEMIA

Causes: Primary hyperparathyroidism, malignancy (PTHrP, bone mets), Vit D toxicity, sarcoidosis, milk-alkali syndrome.

Clinical: "Bones, Stones, Groans, Psychic Moans" — osteitis fibrosa cystica, nephrolithiasis/nephrocalcinosis, constipation/abdominal pain, depression/psychosis.

Severe: Cardiac arrhythmias (short QT)

08

Trace Minerals

MINERAL	KEY BIOCHEMICAL ROLES	DEFICIENCY	TOXICITY / EXCESS	PYQ
Zinc (Zn)	Cofactor for >300 enzymes (carbonic anhydrase, carboxypeptidase, alcohol dehydrogenase, DNA polymerase, superoxide dismutase). Zinc finger proteins (transcription factors). Wound healing. Insulin storage (Zn-insulin hexamer).	Growth retardation, Acrodermatitis enteropathica (perioral/perianal rash), hypogonadism, poor wound healing, impaired taste/smell (hypogeusia/anosmia), immune dysfunction. Night blindness (↓ retinol-binding protein synthesis)	Rare — nausea, copper deficiency (compete for absorption)	2022 (Wilson's)
Copper (Cu)	Ceruloplasmin (ferroxidase — $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$), Cytochrome c oxidase (Complex IV of ETC), Superoxide dismutase (Cu-Zn SOD), Dopamine β-hydroxylase, Lysyl oxidase (collagen cross-linking), Tyrosinase (melanin)	Menkes disease (X-linked): kinky hair (pili torti), neurodegeneration, connective tissue defects, ↓ ceruloplasmin. Acquired: anaemia, bone disease	Wilson's disease: AR, ATP7B gene (copper transport ATPase) → copper accumulates in liver (cirrhosis), brain (neuropsychiatric), eye (Kayser-Fleischer rings in cornea), kidney, joints. ↑ urinary copper, ↓ ceruloplasmin	2022, 2025
Selenium (Se)	Component of Glutathione peroxidase (GPx) — reduces H_2O_2 and lipid peroxides using GSH. Selenocysteine = 21st amino acid (encoded by UGA stop codon with SECIS element). Iodothyronine deiodinase ($\text{T}_4 \rightarrow \text{T}_3$ conversion). Thioredoxin reductase.	Keshan disease (cardiomyopathy, China). Kashin-Beck disease (osteoarthropathy). Impaired antioxidant defence. ↑ cancer risk	Selenosis: garlic odour, hair/nail loss, nausea, neurological symptoms	2014 (antioxidant enzymes)
Iodine (I)	Essential for thyroid hormone synthesis — T_3 (triiodothyronine) + T_4 (thyroxine). Thyroid peroxidase uses I^- to iodinate tyrosine residues on thyroglobulin	Hypothyroidism, Goitre (endemic), Cretinism (congenital — intellectual disability, short stature, deaf-mutism). Most common preventable cause of intellectual disability worldwide	Excess: Jod-Basedow (hyperthyroidism in iodine-replete patient given sudden load)	—
Fluoride (F)	Stabilises hydroxyapatite → fluorapatite (resistant to caries). Used in drinking water fluoridation	Dental caries ↑	Dental fluorosis (mottled enamel), Skeletal fluorosis (bone sclerosis)	—
Manganese (Mn)	Mitochondrial superoxide dismutase (Mn-SOD), Pyruvate carboxylase, Arginase (urea cycle)	Rare — growth retardation, bone abnormalities	Manganism — Parkinson-like syndrome (miners)	—

Chromium (Cr)	Potentiates insulin action (Chromodulin / GTF — Glucose Tolerance Factor)	Impaired glucose tolerance, insulin resistance	Cr(VI) — carcinogen	—
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WILSON'S VS MENKES — COPPER DISORDERS

"Wilson = Too Much copper (Wilts the liver). Menkes = Too Little copper (Kinky hairs, can't Keep copper)"

Wilson's: ATP7B gene (liver/brain). Menkes: ATP7A gene (intestinal export — Cu trapped in gut, can't get OUT).

09 Protein-Energy Malnutrition

FEATURE	MARASMUS	KWASHIORKOR
Definition	Total calorie deficiency (protein + energy both low)	Protein deficiency with relatively adequate calories (carbs present)
Age	Infants < 1 year (early weaning)	Children 1–5 years (post-weaning — starchy diet)
Appearance	"Skin and bones" — emaciated, muscle wasting, no oedema, large head relatively	Pot belly, oedema, "flaky paint" skin, sparse reddish hair (flag sign), moon face
Oedema	Absent	Present (hypoalbuminaemia → ↓ oncotic pressure → pitting oedema)
Serum albumin	Normal or slightly ↓	↓↓ (key feature — <3.5 g/dL)
Serum glucose	Hypoglycaemia	May be normal
Liver	Normal (fat not abundant)	Fatty liver (↓ synthesis of apoproteins → VLDL cannot be assembled → TG accumulates)
Immunity	Severely impaired	Severely impaired (thymus atrophy, ↓ T cells)
Hair changes	Thin, sparse	Hypopigmentation, flag sign (alternating light/dark bands = periods of poor/good nutrition)

KWASHIORKOR — BIOCHEMICAL BASIS OF OEDEMA

★ PYQ 2021, 2023

⚡ The oedema in Kwashiorkor is NOT due to excess sodium or heart failure — it is purely from hypoalbuminaemia reducing plasma oncotic pressure.

► KWASHIORKOR PATHOPHYSIOLOGY

↓ Dietary protein intake

↓ Less amino acids available for liver protein synthesis

↓ Serum albumin synthesis → Hypoalbuminaemia

↓ Starling forces disrupted — oncotic pressure falls

Fluid leaks from capillaries into interstitium → Pitting oedema

↓ Simultaneously — ↓ Apolipoprotein B synthesis

VLDL cannot be assembled → Triglycerides accumulate in liver → Fatty liver

10 Collagen Post-Translational Modifications

⚡ Collagen is the most abundant protein in the body (~30% of total protein). Its synthesis involves an extraordinary number of co- and post-translational modifications — each is a potential exam question.

► COLLAGEN SYNTHESIS — INTRACELLULAR TO EXTRACELLULAR

Step 1 — Gene → Translation mRNA → Pre-pro- α chains (contains signal peptide + N/C propeptides) Gly-X-Y repeat throughout: X = often Pro, Y = often **Hydroxyproline**

↓ Enter ER lumen — Signal peptide cleaved → Pro- α chains

Step 2 — Hydroxylation (ER) — Requires Vitamin C + O_2 + Fe^{2+} **Prolyl hydroxylase:** Proline → **Hydroxyproline** (4-Hyp stabilises triple helix via H-bonds) **Lysyl hydroxylase:** Lysine → **Hydroxyllysine** (attachment point for O-linked glycosylation + cross-link formation) Both use α -ketoglutarate + O_2 ; require Fe^{2+} and **Ascorbate (Vit C)** to re-reduce $Fe^{3+} \rightarrow Fe^{2+}$

↓ Some hydroxyllysines glycosylated with galactose $\pm$ glucose

Step 3 — Triple Helix Formation (ER) Three pro- α chains → C-propeptide nucleates → zipper-like folding → **Procollagen** (triple helix with N/C propeptides)

↓ Secreted via Golgi into extracellular space

Step 4 — N/C Propeptide Cleavage (extracellular) Procollagen N-protease + C-protease cleave propeptides → **Tropocollagen** (native collagen monomer) Cleavage defect → Dermatosparaxis type Ehlers-Danlos (EDS)

↓ Self-assembly into collagen fibrils

Step 5 — Cross-linking by Lysyl Oxidase (extracellular) — requires Copper Lysyl oxidase oxidises Lys/HydroxyLys → Allylsine → forms covalent cross-links between adjacent tropocollagen molecules → **mature collagen fibril** (very high tensile strength) Copper deficiency → defective cross-linking → weak connective tissue (seen in Menkes disease)

COLLAGEN TYPE	LOCATION	DISEASE IF DEFECTIVE
Type I	Bone, skin, tendon, dentin (90% of body collagen)	Osteogenesis Imperfecta (brittle bone disease)
Type II	Cartilage, vitreous humour	Achondrogenesis
Type III	Blood vessels, GI tract, uterus (fetal)	Ehlers–Danlos syndrome (vascular type) → vessel rupture
Type IV	Basement membrane	Alport syndrome (defective Col IV α5 → nephritis + deafness)

11 Viva & Exam Q&A

VITAMINS – FAT-SOLUBLE

Explain the molecular basis of night blindness in Vitamin A deficiency.PYQ 2014-S, 2025

Visual cycle (Wald cycle): 11-cis Retinal (Vit A aldehyde) combines with Scotopsin in rod cells → Rhodopsin. Light → 11-cis Retinal → all-trans Retinal → conformational change → nerve impulse. In dark, all-trans Retinal recycled back to 11-cis Retinal via RPE cells.

In Vit A deficiency: Insufficient 11-cis Retinal for rhodopsin regeneration after bleaching → rod cells fail to respond in dim/dark conditions → nyctalopia (night blindness). Cone cells (colour vision) need Iodopsin and are less sensitive, hence colour vision preserved initially.

Examiner tip: Mention that Vit A is NOT consumed in the visual cycle — it is recycled. Night blindness = defect in RECOVERY phase of the cycle.

Why does Vitamin K deficiency cause bleeding? Name the factors affected.PYQ 2014-S, 2019, 2023

Vit K acts as cofactor for **γ-carboxylase** enzyme. This enzyme adds a carboxyl group (CO₂) to specific glutamate residues in clotting factor precursors → forming **γ-carboxyglutamate (Gla)** residues → these chelate Ca²⁺ → Ca²⁺ binding anchors the factors to phospholipid surface of platelets → essential for clot formation.

Vit K-dependent factors: 2, 7, 9, 10, Protein C, Protein S, Protein Z

Without γ-carboxylation, these factors are produced (PIVKA — Proteins Induced by Vit K Absence) but are non-functional → bleeding tendency → ↑ PT, ↑ PTT.

Warfarin inhibits Vit K epoxide reductase → blocks Vit K recycling → same anticoagulant effect.

HDN (Haemorrhagic disease of newborn): sterile gut → no K2 from bacteria + minimal transfer across placenta + low in breast milk → neonatal bleeding. Prophylactic Vit K given at birth.

 FOLATE TRAP & B12

Explain the folate trap. Why does B12 deficiency cause neurological damage but folate deficiency does not? PYQ 2018, 2022, 2024

Folate trap: In B12 deficiency, the reaction $\text{Methyl-THF} + \text{Homocysteine} \rightarrow \text{THF} + \text{Methionine}$ (by Methionine synthase, needs B12) cannot proceed. All cellular folate becomes trapped as methyl-THF. Cannot be polyglutamated and retained intracellularly. Free THF depleted $\rightarrow$ DNA synthesis impaired $\rightarrow$ megaloblastic anaemia.

Why neuro damage only in B12 deficiency: B12 has a separate second reaction — conversion of $\text{Methylmalonyl-CoA} \rightarrow \text{Succinyl-CoA}$ (needs Adenosylcobalamin). In B12 deficiency, Methylmalonyl-CoA accumulates $\rightarrow$ incorporated into myelin $\rightarrow$ abnormal branched fatty acids $\rightarrow$ **demyelination of dorsal + lateral spinal cord columns** $\rightarrow$ **Subacute Combined Degeneration (SACD)** — posterior column (fine touch, vibration) + lateral corticospinal tract (motor) affected.

Folate deficiency causes anaemia (via same dTMP synthesis block) but has NO second B12 reaction to mimic $\rightarrow$ no SACD.

Danger: Giving folate to B12-deficient patient corrects the anaemia (untraps folate partially?) but doesn't restore B12 $\rightarrow$ SACD continues undetected $\rightarrow$ permanent neurological damage.

What is the significance of methylmalonyl acid in urine for diagnosis of B12 deficiency? PYQ 2024

B12 (as adenosylcobalamin) is required for **Methylmalonyl-CoA mutase**: $\text{Methylmalonyl-CoA} \rightarrow \text{Succinyl-CoA}$. In B12 deficiency, this step fails $\rightarrow$ Methylmalonyl-CoA accumulates $\rightarrow$ excreted as **Methylmalonic acid (MMA)** in urine.

MMA elevation is **specific for B12 deficiency** — not elevated in folate deficiency. This is the key lab test to differentiate the two causes of megaloblastic anaemia. Also elevated in Propionic acidemia and Methylmalonic acidemia (enzyme defects).

Practical: Measure both serum MMA and homocysteine — MMA specific for B12; homocysteine elevated in both B12 and folate deficiency.

IRON & MINERALS

Describe iron absorption, transport and storage. Name the iron-containing proteins. PYQ 2012-S, 2014-S, 2022, 2025

Iron-containing proteins:

- **Haem proteins:** Haemoglobin (O_2 transport), Myoglobin (O_2 storage in muscle), Cytochromes (ETC — I, II, III, IV), Catalase (H_2O_2 decomposition), Peroxidases
- **Non-haem proteins:** Ferritin + Haemosiderin (storage), Transferrin (transport), Ferroportin (export), DMT-1 (intestinal import), Ferredoxin, Iron-Sulphur clusters (Complex I, II)

Absorption: Haem iron $\rightarrow$ HCP1 receptor (25% absorption). Non-haem $Fe^{3+} \rightarrow$ Dcytb (duodenal cytochrome b, needs Vit C to reduce $Fe^{3+} \rightarrow Fe^{2+}$) $\rightarrow$ DMT-1 into enterocyte $\rightarrow$ stored as Ferritin OR exported via Ferroportin + Hephaestin (Cu-dependent ferroxidase oxidises $Fe^{2+} \rightarrow Fe^{3+}$) $\rightarrow$ Transferrin in plasma.

Regulation: Hepcidin (liver) degrades Ferroportin when iron stores high or inflammation $\rightarrow$ blocks export $\rightarrow$ reduces serum iron.

Storage: Ferritin in liver, spleen, bone marrow. Serum ferritin = most sensitive indicator of iron stores. Haemosiderin = insoluble storage, formed from partial lysosomal degradation of ferritin aggregates.

Wilson's disease is a disorder of copper metabolism — justify. PYQ 2022

Wilson's disease is an autosomal recessive disorder caused by mutation in the **ATP7B gene** (chromosome 13q14), encoding a copper-transporting P-type ATPase located in hepatocytes.

Molecular mechanism: ATP7B normally exports copper from hepatocytes into bile for excretion AND incorporates copper into ceruloplasmin. In Wilson's, ATP7B absent/dysfunctional $\rightarrow$ copper not exported into bile $\rightarrow$ not incorporated into ceruloplasmin $\rightarrow$ copper accumulates in liver (toxic, free radical formation) $\rightarrow$ cirrhosis $\rightarrow$ leaks into blood $\rightarrow$ deposits in brain (basal ganglia $\rightarrow$ neuropsychiatric symptoms), cornea (Kayser-Fleischer rings), kidney, joints.

Lab: $\downarrow$ Serum ceruloplasmin (<20 mg/dL), $\uparrow$ urinary copper (>100 μ g/24h), $\uparrow$ liver copper (>250 μ g/g dry weight), $\downarrow$ serum copper (paradoxically — most copper is ceruloplasmin-bound; free/non-ceruloplasmin copper $\uparrow$).

Treatment: D-penicillamine or Trientine (copper chelation); Zinc (competes with copper absorption); Liver transplant in fulminant cases.

 COLLAGEN & PEM

Why does Vitamin C deficiency cause scurvy? What is the molecular link to collagen?PYQ 2018-S, 2021

Molecular link: Collagen synthesis requires hydroxylation of Proline (→ Hydroxyproline) and Lysine (→ Hydroxylysine) by Prolyl hydroxylase and Lysyl hydroxylase respectively. Both enzymes are **mixed-function oxidases** that require Fe^{2+} in their active site. During the reaction, Fe^{2+} is oxidised to Fe^{3+} . **Vitamin C (ascorbate) regenerates Fe^{2+} from Fe^{3+}** — without it, enzyme becomes inactive after one catalytic cycle.

Without Hydroxyproline, the collagen triple helix cannot form properly → unstable procollagen → secreted defective collagen → weak connective tissue → vessel walls fragile → bleeding.

Clinical features explained: Perifollicular haemorrhage (corkscrew hairs) = defective perifollicular capillaries. Bleeding gums = weak gingival connective tissue. Poor wound healing = defective collagen deposition. Subperiosteal bleeding in children = weak periosteum.

Differentiate Marasmus from Kwashiorkor.PYQ 2021, 2023

Marasmus: Total calorie + protein deficiency. Infant < 1 year. Severely emaciated ("skin and bones"), muscle wasting, no subcutaneous fat. No oedema. Serum albumin near normal. Liver normal (body uses fat stores, muscle for energy). No fatty liver. "Marasmus = Metabolic Adaptation" — body adapts to total starvation.

Kwashiorkor: Protein deficiency with adequate calories (carbs). Child 1–5 years. Appearance: oedema (pitting), pot belly, moon face, skin changes (flaky paint dermatosis), hair hypopigmentation + flag sign. Serum albumin ↓↓ (< 3.5 g/dL). Fatty liver (↓ apolipoprotein synthesis → VLDL cannot form → TG accumulates). Oedema = ↓ oncotic pressure from ↓ albumin.

Key biochemical difference: In Kwashiorkor, insulin (from carbs) promotes amino acid uptake into muscle → liver deprived of AAs → ↓ albumin + apolipoprotein synthesis. In Marasmus, amino acids mobilised from muscle for energy + gluconeogenesis → albumin maintained better.

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Essay Question Templates

EQ — IRON ABSORPTION, TRANSPORT, STORAGE [PYQ 2012-S, 2014-S, 2022, 2025]

"Diagrammatically discuss the absorption, transport and storage of iron. Enlist the iron-containing proteins. Justify the role of cytochrome in electron transport chain."

- 1. Dietary Iron:** Two forms — Haem iron (Fe^{2+} , from meat, 25% absorbed, HCP1 receptor) and Non-haem iron (Fe^{3+} , from plants/cereals, lower bioavailability, DMT-1 after reduction to Fe^{2+} by Dcytb/Vit C).
- 2. Absorption (duodenum + upper jejunum):** Fe^{2+} enters enterocyte via DMT-1. Stored as Ferritin (mucosal block) OR exported via Ferroportin. Hephaestin (Cu-containing ferroxidase) oxidises $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ for loading onto Transferrin. Vit C $\uparrow$ absorption by keeping iron in Fe^{2+} form and by $\uparrow$ mucosal cell ferritin synthesis (actually, Vit C mainly reduces Fe^{3+} in lumen).
- 3. Transport:** Transferrin (β_1 -globulin, binds 2 Fe^{3+}) transports iron in plasma. TIBC = Total Iron Binding Capacity. Normal saturation $\sim 33\%$. TfR1 mediates cellular uptake via clathrin-coated endocytosis.
- 4. Storage:** Ferritin (24-subunit protein, stores up to 4500 Fe atoms as ferric oxyhydroxide polymer — soluble, non-toxic). Serum ferritin reflects stores. Haemosiderin (degraded ferritin aggregates, insoluble, Prussian blue stain +) in iron overload.
- 5. Regulation by Hepcidin:** Liver-derived peptide. High iron/inflammation $\rightarrow \uparrow$ Hepcidin $\rightarrow$ degrades Ferroportin $\rightarrow \downarrow$ iron export from enterocytes + macrophages $\rightarrow$ anaemia of chronic disease mechanism.
- 6. Iron-containing proteins:** Haem: Hb, Myoglobin, Cytochromes (b, c, c1, a, a3), Catalase, Peroxidase. Non-haem: Ferritin, Haemosiderin, Transferrin, Fe-S clusters (Complexes I, II, III).

 EQ – FOLATE TRAP [PYQ 2018, 2022, 2024]

"Vitamin B12 should be given along with folic acid to treat folic acid deficient anaemia."

1. **The Problem:** B12 deficiency and folate deficiency both cause megaloblastic anaemia — because both impair dTMP synthesis → defective DNA synthesis → large immature RBCs. Without distinguishing the cause, giving only folate to a B12-deficient patient may seem logical.
2. **The Folate Trap mechanism:** B12 (methylcobalamin) is essential for Methionine Synthase: Methyl-THF + Homocysteine → THF + Methionine. In B12 deficiency, methyl-THF cannot donate its methyl group → all folate trapped as methyl-THF → no free THF → no dTMP synthesis → megaloblastic changes persist.
3. **Why giving folic acid alone to B12-deficient patient is dangerous:** Large doses of folic acid can partially overcome the trap (increase methyl-THF beyond the trapping capacity) → temporarily corrects the blood picture → anaemia improves → patient and doctor think they are treating correctly. BUT the underlying B12 deficiency continues → Methylmalonyl-CoA accumulates → demyelination → SACD progresses silently → irreversible neurological damage.
4. **The correct approach:** ALWAYS diagnose the exact deficiency. Measure serum B12, serum folate, serum MMA (↑ in B12 only), serum homocysteine (↑ in both). Give B12 to all megaloblastic anaemia patients unless folate deficiency is definitively proven.
5. **Why B12 must accompany folate:** In any mixed deficiency (common in malnourished patients), B12 ensures the folate trap is fully released, methionine synthesis is restored (needed for SAM → methylation reactions → myelin maintenance), and the SACD risk is eliminated.

 EQ – CALCIUM HOMEOSTASIS [PYQ 2023, 2016-S]

"How is the homeostasis of plasma calcium level achieved? Indicate factors modifying calcium absorption from gut."

1. **Normal serum calcium:** 8.5–10.5 mg/dL. Three fractions: ionised (50%, biologically active), albumin-bound (40%), complexed with citrate/phosphate (10%). Corrected calcium = measured Ca + $0.8 \times (4 - \text{albumin g/dL})$.
2. **PTH — acute regulator:** ↓ Ca^{2+} → parathyroid glands (CaSR) → PTH secreted → Bone: osteoclast activation → Ca^{2+} + Pi released; Kidney: ↑ tubular Ca^{2+} reabsorption, ↑ Pi excretion, ↑ 1α -hydroxylase → Calcitriol. Net: ↑ serum Ca^{2+} .
3. **Calcitriol (1,25- OH_2 -D3) — chronic regulator:** Intestine: induces Calbindin D → ↑ Ca^{2+} absorption (main route for long-term homeostasis). Bone + kidney: as above. Genomic action via VDR nuclear receptor.
4. **Calcitonin — when Ca^{2+} high:** C-cells of thyroid → inhibits osteoclasts → ↓ bone resorption → ↓ serum Ca^{2+} . Less important in adults; marker for medullary thyroid carcinoma.
5. **Factors increasing Ca^{2+} absorption:** Vit D (calcitriol — main driver), Acidic pH (duodenum), Lactose (forms soluble complex), Lysine/Arginine (amino acids), Low dietary calcium (↑ receptor expression), Growth hormone, PTH (indirectly via calcitriol).
6. **Factors decreasing Ca^{2+} absorption:** Phytates and Oxalates (form insoluble complexes — phytic acid in cereals, oxalic acid in spinach/tea), Fatty acids (form calcium soaps — steatorrhoea → ↓ Ca^{2+}), Alkaline pH, High dietary phosphate, Glucocorticoids (↓ calcitriol synthesis), Excess Mg^{2+} (competes for absorption).

🔥 EXAM DAY HIGH-YIELD QUICKFIRE – MODULE 08

QUESTION	ANSWER
Only stored fat-soluble vitamin (massive reserves)	Vitamin A (liver — 1–2 year supply)
Active form of Vit D	1,25-dihydroxycholecalciferol (Calcitriol) — made in kidney by 1 α -hydroxylase
Storage form of Vit D (measured in blood)	25-hydroxycholecalciferol (Calcidiol) — made in liver
Vit K-dependent clotting factors	II, VII, IX, X + Protein C, S, Z
Why HDN in neonates?	Sterile gut → no K2 from bacteria + low placental transfer + low breast milk Vit K
Vitamin that requires B12 to regenerate	Folate — methyl-THF → THF requires B12 (methionine synthase)
MMA elevation indicates	Vitamin B12 deficiency specifically (not folate)
Pellagra — amino acid connection	Tryptophan → Niacin (60 mg Trp = 1 mg niacin). Defect in Trp metabolism → pellagra in Hartnup disease, carcinoid
PLP-dependent reactions (key 3)	Transamination (ALT/AST), Decarboxylation (DOPA → Dopamine, Glu → GABA), ALA synthase (haem)
Serum ferritin significance	Best indicator of iron stores; also acute phase reactant (↑ in inflammation — can mask deficiency)
Hepcidin mechanism	Liver peptide → binds Ferroportin → internalisation → ↓ iron export from gut + macrophages → ↓ serum iron
Copper-dependent enzymes (key 4)	Ceruloplasmin (ferroxidase), Cytochrome c oxidase (Complex IV), Cu-Zn SOD, Lysyl oxidase (collagen cross-linking)
Selenium-dependent enzyme	Glutathione peroxidase (GPx) — reduces H ₂ O ₂ using GSH
Vitamin C — two enzymatic roles in collagen	Cofactor for Prolyl hydroxylase (→ Hyp) AND Lysyl hydroxylase (→ HyLys); both need Vit C to regenerate Fe ²⁺
Kwashiorkor oedema mechanism	↓ Dietary protein → ↓ Albumin synthesis → ↓ plasma oncotic pressure → fluid into interstitium

MODULE 08 MASTER MNEMONICS

🟡 FAT-SOLUBLE VITAMINS

"Fat Dads Eat Kale" = A, D, E, K

🟡 VIT K FACTORS

"1972 PC" = II, VII, IX, X + Protein C (& S, Z)

● PELLAGRA 4DS

**"Dermatitis, Diarrhoea, Dementia,
Death"**

● CALCIUM ABSORPTION INHIBITORS

**"Phytates + Oxalates = PO boxes
block Ca^{2+} "**

● WILSON'S VS MENKES

**"Wilson Wilts the liver (excess Cu).
Menkes can't Keep Cu
(deficiency)"**

● HYPERCALCAEMIA

**"Bones, Stones, Groans, Psychic
Moans"**

MODULE 08 · NUTRITION, VITAMINS & MINERALS

Caffeine & Cadaver · MBBS Biochemistry Notes · 2010-2025

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